Harshana Lakshara Fernando

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SUMMARY

Full-Stack Software Engineer experienced in end-to-end application delivery, legacy integrations, and production support. Strong in **Java**, **Spring Boot**, **Kafka**, **React**, and DevOps workflows. Known for rapid learning, taking ownership of distributed systems, and choosing solutions that balance reliability, scalability, and real product needs.

EDUCATION

B.Sc. (Hons) in Engineering — Computer Science and Engineering, University of Moratuwa 2022 – 2026
G.C.E. A/L: 3 A's, Physical Science Stream, District Rank 17

EXPERIENCE

Software Engineer — Trading Systems Sep 2026 – Present
CAL Colombo, Sri Lanka

- Sole engineer on CAL's in-house electronic trading platform: a direct gateway to the **Colombo Stock Exchange (CSE)** plus trading screens and ops tools, replacing a vendor-built broker system.
- Carried **CSE certification** to the final stage: all application-level cases of the **188** exchange test cases re-run on CSE's test system with full exchange-message logs.
- Built an **event-sourced, CQRS** engine with a **Chronicle Queue** write-ahead log on the order path and **PostgreSQL** as an async read model, so database writes never block execution.
- Implemented **FIX 5.0 SP2** order routing and drop copy (**QuickFIX/J**) and a custom **Java NIO** decoder for the **MITCH** UDP market-data feed; built a Certification Runner that diffs every field against CSE's expected results.

R&D Software Engineer — KōreroCare Platform Mar 2026 – Aug 2026
Empathic Computing Laboratory, University of Auckland Remote

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.

Co-founder & Lead Engineer (Freelance) Jul 2025 – Aug 2026
ScaleX — SaaS Development Lab Remote / Sri Lanka

- Co-founded ScaleX and owned the end-to-end lifecycle of multiple SaaS MVPs, from architecture through cloud-native deployment.
- Built **AgriNova** (**Next.js**, **Prisma**, **Supabase**/**PostgreSQL**), an inventory and commission platform across 5 branches and 329+ SKUs, tracking 194K+ units and 860+ invoices (Rs. 64.4M+).
- Cut repeat dashboard data-fetch latency from ∼384ms to <1ms with **React Query** caching and pagination.

Software Engineer Intern — ISA Aviation Technology Dec 2024 – Jun 2025
Information Systems Associates (Pvt) Ltd Colombo, Sri Lanka

- Owned design and delivery of a distributed **Spring Boot**/**Kafka**-backed PNR ID generation service, **live in production**, achieving **25+ PNRs/sec** at **<100ms latency**.
- Designed a **pool-based PNR allocation model** replacing legacy Oracle-sequence generation, with **concurrency-safe** Kafka-backed allocation and Java **ReentrantLock** critical sections.
- Led the rollout to AeroOrder using database-driven **feature toggles** and a **strangler-pattern** migration, bridging the **Java 8** AeroMart monolith via **gRPC** clients and Kafka consumers.
- Resolved production **Kafka consumer lag** and offset-reset issues with graceful shutdown hooks, retry handling, validation, and a Producer–Consumer refactor.
- Built load testing and monitoring with **JMeter**, **Prometheus**, and **Grafana**; maintained **>90% unit test coverage**.

OPEN SOURCE

Merged: kgateway #14851 (CNCF) — `kgateway_build_info` metric, shipped in v2.5.0-beta.2; kubernetes-mcp-server #1482 and #1549 — test-startup fix, RBAC metadata test.
In review: Kubeflow Spark Operator #3249 (100% executor-metrics coverage, LGTM); KEDA #8317 (CNCF, `onlyReadyPods` workload-scaler option).

PROJECTS

Supply Chain DApp Solidity, Next.js, Polygon

- Developed a full-stack supply chain tracker with trustless escrow.
- Technologies: Solidity, Next.js

Governance DAO Solidity, Hardhat

- Engineered a decentralized autonomous organization for voting.
- Technologies: Solidity, OpenZeppelin

ZK-Rollups Scalability Circom, Solidity

- Researched batching and throughput enhancement on Ethereum Layer 2.
- Technologies: Circom, Solidity

Fintech & Crypto Lab. Research

- Group project focusing on decentralized finance mechanisms.
- Technologies: DeFi, Blockchain

SKILLS

Backend: Java, **Spring Boot**, Go/Gin, Python, Node.js, REST, **gRPC**, GraphQL, Hibernate
Frontend: React, Next.js, **TypeScript**, JavaScript, React Query
Messaging & APIs: Kafka, RabbitMQ, WebSockets, Protobuf, JWT, OAuth2/OIDC
Data: PostgreSQL, MySQL, Oracle DB, Redis, Couchbase, DynamoDB, Prisma
Cloud / DevOps: AWS, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Linux
Observability / Testing: Prometheus, Grafana, JMeter, Gatling, JUnit, Mockito, Pytest